3512. Minimum Operations to Make Array Sum Divisible by K

Difficulty: Easy

Link: https://leetcode.com/problems/minimum-operations-to-make-array-sum-divisible-by-k/description/?envType=daily-question&envId=2025-11-29

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Problem:

You are given an integer array `nums` and an integer `k`. You can perform the following operation any number of times:

- Select an index `i` and replace `nums[i]` with `nums[i] - 1`.

Return the **minimum** number of operations required to make the sum of the array divisible by `k`.

Example 1:

Input: nums = [3,9,7], k = 5

Output: 4

Explanation:

- Perform 4 operations on `nums[1] = 9`. Now, `nums = [3, 5, 7]`.
- The sum is 15, which is divisible by 5.

Example 2:

Input: nums = [4,1,3], k = 4

Output: 0

Explanation:

- The sum is 8, which is already divisible by 4. Hence, no operations are needed.

Example 3:

Input: nums = [3,2], k = 6

Output: 5

Explanation:

- Perform 3 operations on `nums[0] = 3` and 2 operations on `nums[1] = 2`. Now, `nums = [0, 0]`.
- The sum is 0, which is divisible by 6.

Constraints:

- `1 <= nums.length <= 1000`
- `1 <= nums[i] <= 1000`
- `1 <= k <= 100`